$\Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \cdots$

$= \Sigma^0 \cup \Sigma^+$

$= \{\epsilon\} \cup \Sigma^+$

$= \Sigma^+ \cup \{\epsilon\}$

Concatenation of strings:

Let x and y be strings. Then xy denotes the concatenation of x and y, that is, the string formed by making a copy of x and following it by a copy of y.

Example: x = "my"

y = "computer"

xy = "mycomputer"

yx = "computermy"

Languages:

A set of strings all of which are chosen from some Σ^*, where Σ is a particular alphabet, is called a language.

If Σ is an alphabet, and $L \subseteq \Sigma^*$, then L is a language over Σ

Example:

alphabet, $\Sigma^* = \{a, b, c, \cdots, z\}$, $\Sigma^0 = \{\epsilon\}$

$\Sigma^1 = \{a\}$

$\Sigma^2 = \{am, an, to, of, \cdots\}$

$\Sigma^3 = \{boy, eat, cat, sat, mat, \cdots\}$

$\vdots$

$\therefore \Sigma^* = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \Sigma^3 \cup \cdots$

$= \{\epsilon\} \cup \{a\} \cup \{am, an, to, of, \cdots\} \cup \{boy, eat, cat,$

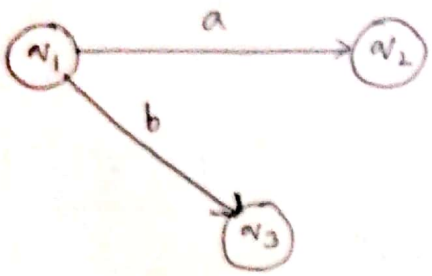
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$= \{ \epsilon, a, am, an, to, of, \ldots, boy, eat, cat, sat, mat, \ldots \}$

A set of strings all of which are chosen from Σ^* is called English language.

Deterministic Finite Automata

Deterministic refers to the fact that on each input there is one and only one state to which the automation can transition from its current state.



Definition of DFA (Deterministic Finite Automata)

1. Finite set of states, Q
2. Input symbols Σ
3. Transition function δ

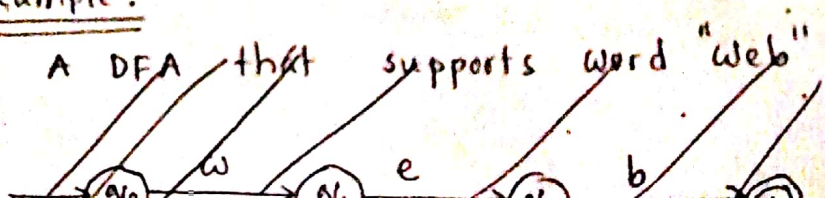
$$\delta(q_1, a) = q_2 \quad \text{[returns one state]}$$

4. starting state q_0.
5. A set of final or accepting states F.

$$\therefore \text{DFA} = (Q, \Sigma, \delta, q_0, F)$$

Example:

A DFA that supports word "web"



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